

Sri Lankan Black Pepper

Pepper Variety Knowledge Document

Knowledge Domain: 01 — Planting Practices (Approved Varieties)

Project: RAG Chat System for Sri Lankan Black Pepper Knowledge

1. Overview

This document provides structured knowledge on all currently recommended black pepper (*Piper nigrum*) varieties for cultivation in Sri Lanka. It covers two internationally introduced commercial varieties, one local DEA-recognised selection, and three newly released DEA hybrid varieties — totalling six active varieties.

2. All Recommended Varieties — Summary Table

The following six varieties are currently endorsed for black pepper cultivation in Sri Lanka by the Department of Export Agriculture (DEA).

Variety	Origin	Specialities	Suitability	Soil Type	Spacing (m)	Vines/ha	Pit (cm)
Panniyur-1	Introduced — India (1970s)	World's first pepper hybrid; long spikes and very high yield. Parent of all three DEA hybrid varieties.	Vigorous grower that thrives in open areas with high humidity. Best for low to mid-country wet zones.	Prefers well-drained loamy soil.	2.5 x 2.5	1600	60 x 60 x 60
Kuching	Introduced — Malaysia (1970s)	High piperine content and excellent dry-to-green recovery ratio. Most adaptable to adverse conditions.	Tolerates drier zones and adverse soil conditions. Recommended across wet, intermediate, and dry zones.	Thrives in sandy loam or well-drained gravelly soil.	2.0 x 2.0	2500	50 x 50 x 50

Variety	Origin	Specialities	Suitability	Soil Type	Spacing (m)	Vines/ha	Pit (cm)
GK 49	Local Selection — Sri Lanka (DEA)	Sri Lanka's premier high-yielding local selection with superior pungency and quality. Replaces the generic 'Local' variety classification.	Highly adaptable to Sri Lankan agro-climatic conditions. Performs best in Red-Yellow Podzolic soils of wet and intermediate zones.	Adapts well to Red-Yellow Podzolic, loamy, and clay-loam soils.	2.5 x 2.5	1600	60 x 60 x 60
Dingi Rala	DEA Hybrid — Sri Lanka (Panniyur-1 × GK 49)	Balanced high-performance hybrid combining the yield of Panniyur-1 with the pungency of GK 49.	Well-suited to wet and intermediate zones at low to mid-country elevations.	Performs best in well-drained loamy and Red-Yellow Podzolic soils.	2.5 x 2.5	1600	60 x 60 x 60
Bootawe Rala	DEA Hybrid — Sri Lanka (Panniyur-1 × DM 7)	Highest yielding of all recommended varieties at 2,724 g/vine/year. Longest panicle at 14 cm. Highest piperine content at 6.3%.	Best suited to intermediate and up-country zones. Recommended replacement for IW 5 in mid to up-country districts.	Well-drained loamy, clay-loam, and Red-Yellow Podzolic soils.	2.5 x 2.5	1600	60 x 60 x 60
Kohukumbure Rala	DEA Hybrid — Sri Lanka (MW 21 × Panniyur-1)	Highest oleoresin (15.4%) and oil content (3.6%) of all varieties — premium export-quality pepper. Heaviest berry weight per litre at 613.43 g.	Best suited to mid to up-country wet zones. Recommended for districts requiring export-grade quality output.	Well-drained loamy and Red-Yellow Podzolic soils in wetter, cooler elevations.	2.5 x 2.5	1600	60 x 60 x 60

3. Introduced Commercial Varieties

Panniyur-1 and Kuching were introduced to Sri Lanka in the 1970s and remain the most widely planted commercial varieties.

3.1 Panniyur-1

Origin	Introduced — India (1970s)
Specialities	World's first pepper hybrid; long spikes and very high yield. Parent of all three DEA hybrid varieties.
Suitability	Vigorous grower that thrives in open areas with high humidity. Best for low to mid-country wet zones.
Soil Recommendation	Prefers well-drained loamy soil.
Planting Spacing	2.5 x 2.5 meters
Vine Density	1600 vines per hectare
Pit Dimensions	60 x 60 x 60 cm

3.2 Kuching

Origin	Introduced — Malaysia (1970s)
Specialities	High piperine content and excellent dry-to-green recovery ratio. Most adaptable to adverse conditions.
Suitability	Tolerates drier zones and adverse soil conditions. Recommended across wet, intermediate, and dry zones.
Soil Recommendation	Thrives in sandy loam or well-drained gravelly soil.
Planting Spacing	2.0 x 2.0 meters
Vine Density	2500 vines per hectare
Pit Dimensions	50 x 50 x 50 cm

4. Local Selection — GK 49

GK 49 is the Department of Export Agriculture's officially recognised premier local selection of black pepper in Sri Lanka. It is a high-yielding, superior-quality variety that replaces the previously used generic 'Local' classification. GK 49 is widely grown by pepper cultivators across wet and intermediate zones.

4.1 GK 49

Origin	Local Selection — Sri Lanka (DEA)
Specialities	Sri Lanka's premier high-yielding local selection with superior pungency and quality. Replaces the generic 'Local' variety classification.
Suitability	Highly adaptable to Sri Lankan agro-climatic conditions. Performs best in Red-Yellow Podzolic soils of wet and intermediate zones.
Soil Recommendation	Adapts well to Red-Yellow Podzolic, loamy, and clay-loam soils.
Planting Spacing	2.5 x 2.5 meters
Vine Density	1600 vines per hectare
Pit Dimensions	60 x 60 x 60 cm

5. DEA Hybrid Varieties (New Releases)

The Department of Export Agriculture has recently introduced three new hybrid black pepper varieties developed from crosses involving Panniyur-1. These hybrids offer superior yield, pungency, and export-quality metrics, and replace IW 5 in mid and up-country district recommendations.

Hybrid Technical Comparison:

Specification	Dingi Rala	Bootawe Rala	Kohukumbure Rala
Parent Varieties	Panniyur-1 × GK 49	Panniyur-1 × DM 7	MW 21 × Panniyur-1

Specification	Dingi Rala	Bootawe Rala	Kohukumbure Rala
Panicle Length	12 cm	14 cm ★ Longest	12 cm
Filling Percentage	80%	80%	80%
Yield (g/vine/year)	2,245 g	2,724 g ★ Highest	2,340 g
Dry Weight Ratio	327:1	327:1	246:1
Oleoresin %	12.9%	12.9%	15.4% ★ Highest
Oil %	2.8%	3.1%	3.6% ★ Highest
Piperine %	5.6%	6.3% ★ Highest	6.0%
Weight of 1 Litre	571.86 g	490.88 g	613.43 g ★ Heaviest
Weight of 1,000 Seeds	63.44 g	47.19 g	53.31 g
Best For	Balanced yield & quality	Max yield & piperine	Export quality — oil & oleoresin

Selection Guide: Choose *Bootawe Rala* for maximum yield (2,724 g/vine/yr) and highest piperine (6.3%). Choose *Kohukumbure Rala* for premium export quality (highest oleoresin 15.4%, oil 3.6%). Choose *Dingi Rala* as a balanced, high-performing hybrid across wet and intermediate zones.

5.1 Dingi Rala

Origin	DEA Hybrid — Sri Lanka (Panniyur-1 × GK 49)
Specialities	Balanced high-performance hybrid combining the yield of Panniyur-1 with the pungency of GK 49.
Suitability	Well-suited to wet and intermediate zones at low to mid-country elevations.
Soil Recommendation	Performs best in well-drained loamy and Red-Yellow Podzolic soils.

Planting Spacing	2.5 x 2.5 meters
Vine Density	1600 vines per hectare
Pit Dimensions	60 x 60 x 60 cm
Panicle Length	12 cm
Filling %	80%
Yield (g/vine/year)	2,245 g/year
Dry Weight Ratio	327:1
Oleoresin %	12.9%
Oil %	2.8%
Piperine %	5.6%
Weight per Litre	571.86 g
Weight per 1,000 Seeds	63.44 g

5.2 Bootawe Rala

Origin	DEA Hybrid — Sri Lanka (Panniyur-1 × DM 7)
Specialities	Highest yielding of all recommended varieties at 2,724 g/vine/year. Longest panicle at 14 cm. Highest piperine content at 6.3%.
Suitability	Best suited to intermediate and up-country zones. Recommended replacement for IW 5 in mid to up-country districts.
Soil Recommendation	Well-drained loamy, clay-loam, and Red-Yellow Podzolic soils.
Planting Spacing	2.5 x 2.5 meters
Vine Density	1600 vines per hectare
Pit Dimensions	60 x 60 x 60 cm
Panicle Length	14 cm

Filling %	80%
Yield (g/vine/year)	2,724 g/year ★ Highest
Dry Weight Ratio	327:1
Oleoresin %	12.9%
Oil %	3.1%
Piperine %	6.3% ★ Highest
Weight per Litre	490.88 g
Weight per 1,000 Seeds	47.19 g

5.3 Kohukumbure Rala

Origin	DEA Hybrid — Sri Lanka (MW 21 × Panniyur-1)
Specialities	Highest oleoresin (15.4%) and oil content (3.6%) of all varieties — premium export-quality pepper. Heaviest berry weight per litre at 613.43 g.
Suitability	Best suited to mid to up-country wet zones. Recommended for districts requiring export-grade quality output.
Soil Recommendation	Well-drained loamy and Red-Yellow Podzolic soils in wetter, cooler elevations.
Planting Spacing	2.5 x 2.5 meters
Vine Density	1600 vines per hectare
Pit Dimensions	60 x 60 x 60 cm
Panicle Length	12 cm
Filling %	80%
Yield (g/vine/year)	2,340 g/year

Dry Weight Ratio	246:1
Oleoresin %	15.4% ★ Highest
Oil %	3.6% ★ Highest
Piperine %	6.0%
Weight per Litre	613.43 g ★ Heaviest
Weight per 1,000 Seeds	53.31 g